# The Transmission of U.S. Monetary Policy Shocks across Volatility Regimes

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# 1 Introduction

This project studies how U.S. monetary policy shocks affect the economy and financial markets, and whether the effects differ across market conditions. In particular, we compare the transmission of a monetary tightening when volatility is high and when volatility is low.

We build a monthly VAR with a small set of variables including industrial production growth, inflation, the Federal Funds Rate, a credit spread (BAA–AAA), and S&P 500 returns. The main expected result is a state-dependent transmission: under high volatility, a tightening shock should lead to a stronger and more persistent increase in credit spreads and a more negative equity response compared to low volatility. In periods of high volatility, often associated with financial uncertainty, risk premiums and credit spreads are more sensitive, amplifying the transmission of monetary policy.

Identification of the monetary policy shock is implemented in two ways. First, we use a recursive Cholesky identification as a baseline, with an ordering consistent with monthly data: macro variables are treated as relatively slow-moving, the policy rate comes next, and financial variables react contemporaneously to policy surprises Christiano 1998. Second, we use sign restrictions, imposing intuitive signs for a tightening shock following the agnostic approach Uhlig 2005. For both identification strategies, we compute impulse response functions (IRFs) and construct bootstrap confidence bands, separately for low and high volatility.

# 2 Methodology

## 2.1 Data

We use monthly U.S. data and build a small set of variables that captures the main channels through which monetary policy affects the economy and financial markets:

- Industrial Production growth is our proxy for real economic activity
- Inflation (CPI-based) measures changes in the general price level
- Federal Funds Rate (policy rate) is the main monetary policy instrument in the U.S. In the VAR, a positive shock to this variable is interpreted as a monetary tightening
- Credit spread (BAA–AAA) measures stress in corporate credit markets. A higher spread means tighter credit conditions and higher compensation required for holding riskier corporate debt
- S&P 500 return captures equity market performance

The key idea is that the same monetary policy shock can have different effects depending on market conditions. This is why we estimate the model separately for low-volatility and high-volatility months.

## 2.2 VIX Proxy

Since the VIX was created in 1990 and our dataset starts from 1962 we use a proxy constructs as a monthly volatility proxy based on the realized volatility of the S&P 500, calculated as the annualized standard deviation of daily log-returns:

$$\sigma_m^{RV} = \sqrt{\frac{1}{N-1}\sum_{t=1}^{N}(r_t - \bar{r})^2} \times \sqrt{252} \times 100$$

To ensure consistency with the VIX, we apply a calibration factor $\phi$ for the variance risk premium:

$$\widehat{VIX}_m = \sigma_m^{RV} \times \phi$$

where $\phi$ is the mean ratio between the official VIX and the realized volatility during the 1990-2020.

Then we create a volatility dummy based on the $75^{th}$ percentile ($\tau$) of the calibrated proxy:
- High Volatility Regime ($D = 1$): $\widehat{VIX}_m > \tau$
- Low Volatility Regime ($D = 0$): $\widehat{VIX}_m \leq \tau$

## 2.3 VAR identification

In our project the monthly VAR variable is

$$y_t = \begin{bmatrix} IP_growth_t \\ Inflation_t \\ FFR_t \\ Spread_t \\ SP500_ret_t \end{bmatrix}$$

where $IP_growth_t$ and $Inflation_t$ are monthly log-growth rates (in %), $FFR_t$ is the Federal Funds Rate in levels, $Spread_t$ is the credit spread (BAA–AAA) in levels, and $SP500_ret_t$ is the monthly equity log-return (in %).
For each regime (LOW volatility and HIGH volatility) we estimate a VAR($p$):

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma_u.$$

### Variables order

Following the recursive identification strategy of Christiano, Eichenbaum, and Evans (1998), we assume a recursive structure where macroeconomic variables (Industrial Production and Inflation) are slow moving variable and do not respond contemporaneously to policy shocks for this reason the Federal Funds Rate is ordered after these variables. Finally, financial variables (Credit Spread and SP 500) are ordered last, as they are 'fast-moving' and react instantaneously to monetary policy surprises.

### Lag identification

We pick $p$ once on the full sample using AIC and then keep the same $p$ for both regimes to compare them, and the lag that minimizes the AIC corresponds to p=10

| Lag | AIC Value |
|---|---|
| 1 | -6.1965 |
| 2 | -6.5193 |
| 3 | -6.5968 |
| 4 | -6.6133 |
| 5 | -6.6176 |
| 6 | -6.6140 |
| 7 | -6.6436 |
| 8 | -6.6836 |
| 9 | -6.7161 |
| **10** | **-6.7355** |
| 11 | -6.7182 |
| 12 | -6.7077 |

The reduced-form VAR residuals $u_t$ are generally correlated across equations, so they do not have a direct economic interpretation. To interpret a *structural* shock (e.g., a monetary policy tightening), we map reduced-form shocks into structural shocks via

$$u_t = B_0 \varepsilon_t, \qquad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I, \qquad B_0 B_0' = \Sigma_u,$$

where $\Sigma_u = \mathbb{E}[u_t u_t']$ is the covariance matrix of reduced-form residuals. Different identification schemes correspond to different valid choices of $B_0$ satisfying $B_0 B_0' = \Sigma_u$.

## 2.4 Cholesky identification

Let $S$ be the lower-triangular Cholesky factor of $\Sigma_u$:

$$\Sigma_u = SS', \qquad u_t = S\eta_t, \qquad \eta_t \sim (0, I).$$

With this identification, the ordering of variables in $y_t$ matters. In our ordering (macro variables first, policy next, markets last), financial variables can react immediately to a policy shock, which is consistent with the idea that markets adjust quickly to monetary policy news.
If the VAR admits the moving-average representation $y_t = \mu + \sum_{h \geq 0} \Psi_h u_{t-h}$, then the orthogonalized impulse responses under Cholesky are

$$IRF(h) = \Psi_h S.$$

## 2.5 Zero–sign restrictions

Cholesky is simple but relies on a specific recursive ordering. Starting from the Cholesky factorization $\Sigma_u = SS'$, any orthonormal matrix $H$ with $H'H = I$ generates a valid structural impact matrix

$$B_0(H) = SH,$$

because $B_0(H)B_0(H)' = SHH'S' = SS' = \Sigma_u$. The corresponding impulse responses are

$$IRF(h; H) = \Psi_h SH.$$

### Zero restrictions

In our project we treat $IP_growth$ and $Inflation$ as slow monving variables. The zero restriction we impose is: at impact of the identified monetary policy shock, to compute this we impose this structurally by restricting the rotation to be block-diagonal:

$$H = \begin{bmatrix} I_2 & 0 \\ 0 & Q \end{bmatrix}, \qquad Q'Q = I.$$

This means we only rotate shocks within the ($FFR$, $Spread$, $SP500$) and do not mix them contemporaneously with the first two variables. As a result, the impact responses of $IP_growth$ and $Inflation$ to the identified shock are zero by construction.

**Sign restrictions**

After imposing the *zero timing* structure, we only rotate shocks inside the block (policy and financial variables). In practice, we use a block-diagonal rotation matrix

$$H = \text{diag}(I_2, Q),$$

We compute sign restrictions according the economic sense of a monetary policy shock:
- $FFR$ increases (+)
- $Spread$ increases (+)
- $SP500$ return decreases (-)

Example of a block rotation matrix.

$$H = \begin{bmatrix} 1.0000 & 0 & 0 & 0 & 0 \\ 0 & 1.0000 & 0 & 0 & 0 \\ 0 & 0 & 0.9747 & 0.1637 & 0.1521 \\ 0 & 0 & 0.0709 & 0.4189 & -0.9052 \\ 0 & 0 & -0.2119 & 0.8931 & 0.3967 \end{bmatrix}$$

**Accept/reject algorithm**

The procedure is implemented as:

1. Estimate the VAR and compute $\Sigma_u$.
2. Compute $S = \text{chol}(\Sigma_u)$.
3. Draw random orthonormal matrices $Q$ and build $H = \text{diag}(I_2, Q)$.
4. Compute $IRF(h; H) = \Psi_h SH$.
5. Keep draws that satisfy the sign conditions on $(FFR, Spread, SP500)$ at the chosen horizon(s).

We summarize IRFs across accepted draws (e.g., median) and report confidence bands.

## 2.6 Confidence bands

For each identification scheme (Cholesky and zero–sign restrictions), we compute confidence bands using a bootstrap procedure.

1. generate many bootstrap samples from the estimated VAR,
2. re-estimate the VAR on each bootstrap sample,
3. re-identify the structural shock and recompute the IRFs,
4. take empirical percentiles across the bootstrap IRFs (16–84).

We run the bootstrap separately in the LOW and HIGH volatility subsamples, so the uncertainty can differ across regimes.

## 2.7 FEVD

FEVD summarizes which shocks matter most for each variable at a given forecast horizon. In our setting, for a horizon $H$, FEVD answers: *what fraction of the $H$-step-ahead forecast error variance of variable $i$ is explained by shock $j$?*

Using structural impulse responses, the contribution of shock $j$ to variable $i$ is based on the cumulative squared responses up to horizon $H$:

$$\mathrm{Num}_{i,j}(H) = \sum_{h=0}^{H} IRF_{i,j}(h)^2.$$

The FEVD is obtained by dividing this numerator by the total forecast error variance of variable $i$ at the same horizon.

We should expect the clearest differences across regimes to show up in the financial variables. In high-volatility months, the "policy/financial" shocks typically explain a larger share of the forecast error variance of the credit spread (BAA–AAA) and SP500 returns, because markets react more strongly when risk appetite is low and stress is high. For macro variables like IP growth and inflation, the FEVD is often dominated by their own shocks and tends to change less across regimes, or to be estimated with more uncertainty.

# 3 Empirics

## 3.1 Cholesky and Zero–Sign restrictions

Figure 1 reports the full set of impulse responses under the recursive (Cholesky) identification, separately for LOW volatility (blue) and HIGH volatility (red), with bootstrap bands.

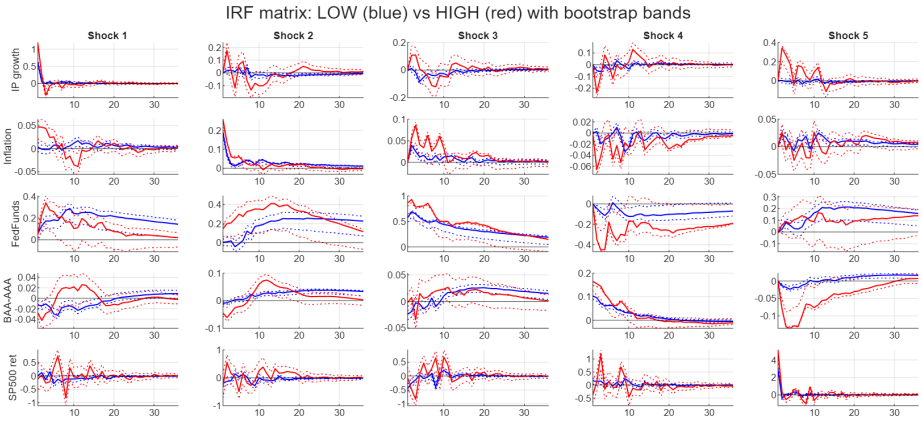



Figure 1: Cholesky IRF matrix

The column corresponding to the monetary policy shock is the third where the FedFunds response is clearly positive on impact and then gradually decays toward zero. Along this shock:

- **FedFunds** increases immediately and shows persistence over several months. The response is broadly similar across regimes, but the HIGH-volatility bands are wider.
- **Credit spread (BAA–AAA)** increases after tightening, consistent with tighter credit conditions. The increase is visibly larger in the HIGH-volatility regime and tends to last longer.
- **S&P500 returns** drop on impact and then revert toward zero relatively quickly. The response is more volatile and less precisely estimated in the HIGH-volatility regime (wider bands).
- **IP growth** and **Inflation** react more slowly and with smaller magnitudes. Their short-run responses are close to zero, and uncertainty is substantially higher in the HIGH-volatility subsample.

Figure 2 reports IRFs to the monetary policy tightening shock identified using the zero–sign approach: IP growth and inflation are restricted not to react contemporaneously (timing/zero restriction), and the shock is selected so that FedFunds rises, the spread widens, and S&P500 returns fall at impact (sign restrictions on the fast block).

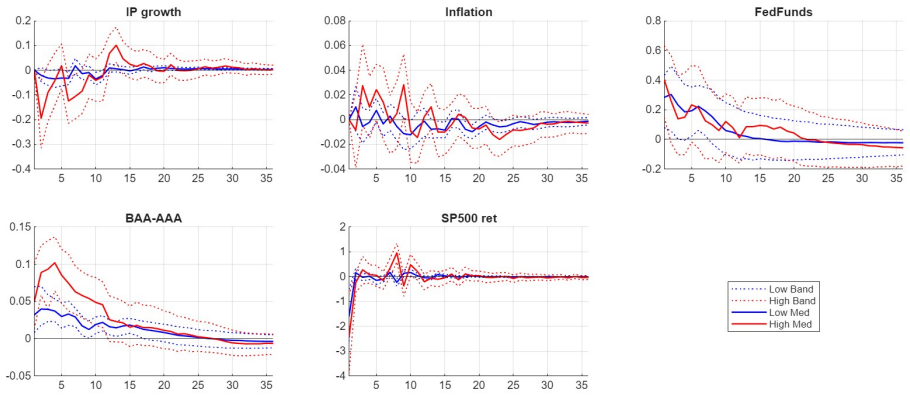



Figure 2: Zero–sign IRF

The resulting responses are economically clear:

- **FedFunds:** positive impact response with gradual mean reversion. The HIGH-volatility response is slightly more uncertain (wider bands).
- **BAA–AAA:** strong positive response, especially in HIGH volatility. The red median is noticeably above the blue median in the first year, indicating a stronger tightening of credit conditions in stressed periods. La Fed alza i tassi, ma il costo del prestito per le imprese sale molto di più del tasso Fed perché aumenta il premio per il rischio.
- **S&P500 returns:** immediate negative impact response, with faster reversion toward zero. The impact drop is larger and more volatile in HIGH volatility (red), consistent with stronger market sensitivity in stress periods.
- **IP growth and Inflation:** responses remain relatively small and are estimated with wide uncertainty, particularly in HIGH volatility. Compared to financial variables, the macro effects are slower and less precisely pinned down.

The key result is that the transmission of monetary tightening is stronger through **financial conditions** when volatility is high. In particular, the credit spread response is larger and more persistent in the HIGH-volatility regime, while uncertainty bands are also wider. This pattern is consistent with the idea that in stressed markets a policy tightening leads to a stronger repricing of risk and tighter credit, while macro variables adjust more gradually.

## 3.2 FEVD

Tables 1,2,3 report the FEVD at 12, 24, and 36 months for the LOW- and HIGH-volatility regimes. In the LOW-volatility regime, each variable is largely explained by its own shock (high diagonal entries)

In the HIGH-volatility regime, the decomposition becomes more financial-driven. The credit spread (BAA–AAA): the share explained by its own shock drops sharply (from about 0.79 at 12 months in LOW to about 0.44 in HIGH), while the share explained by equity shocks (SP500 returns) becomes large (around 0.42 at 12 months). This pattern remains visible at longer horizons (24–36 months), where equity shocks continue to explain a sizeable fraction of the spread variance in HIGH volatility.

Overall, the FEVD tables support when volatility is high, monetary/financial disturbances propagate more through risk premia and credit conditions, and these channels account for a larger share of future uncertainty in spreads and (to a lesser extent) real activity.

Table 1: FEVD at 12 months (LOW vs HIGH)

**LOW**

| | IPGrowth | Inflation | FedFunds | BAA_AAA | SP500Ret |
|---|---|---|---|---|---|
| IP growth | 0.91 | 0.02 | 0.04 | 0.03 | 0.01 |
| Inflation | 0.01 | 0.84 | 0.06 | 0.04 | 0.04 |
| FedFunds | 0.12 | 0.04 | 0.78 | 0.02 | 0.03 |
| BAA–AAA | 0.08 | 0.05 | 0.04 | 0.79 | 0.04 |
| SP500 ret | 0.03 | 0.02 | 0.05 | 0.02 | 0.88 |

**HIGH**

| | IPGrowth | Inflation | FedFunds | BAA_AAA | SP500Ret |
|---|---|---|---|---|---|
| IP growth | 0.73 | 0.05 | 0.03 | 0.05 | 0.14 |
| Inflation | 0.07 | 0.66 | 0.16 | 0.09 | 0.02 |
| FedFunds | 0.06 | 0.12 | 0.68 | 0.12 | 0.02 |
| BAA–AAA | 0.03 | 0.10 | 0.02 | 0.44 | 0.42 |
| SP500 ret | 0.04 | 0.03 | 0.06 | 0.07 | 0.80 |

Table 2: FEVD at 24 months (LOW vs HIGH)

**LOW**

| | IPGrowth | Inflation | FedFunds | BAA_AAA | SP500Ret |
|---|---|---|---|---|---|
| IP growth | 0.93 | 0.02 | 0.04 | 0.03 | 0.01 |
| Inflation | 0.02 | 0.82 | 0.06 | 0.04 | 0.06 |
| FedFunds | 0.15 | 0.12 | 0.61 | 0.03 | 0.09 |
| BAA–AAA | 0.07 | 0.21 | 0.11 | 0.57 | 0.04 |
| SP500 ret | 0.03 | 0.03 | 0.05 | 0.02 | 0.87 |

**HIGH**

| | IPGrowth | Inflation | FedFunds | BAA_AAA | SP500Ret |
|---|---|---|---|---|---|
| IP growth | 0.72 | 0.05 | 0.03 | 0.05 | 0.14 |
| Inflation | 0.07 | 0.63 | 0.16 | 0.11 | 0.03 |
| FedFunds | 0.05 | 0.19 | 0.61 | 0.13 | 0.02 |
| BAA–AAA | 0.03 | 0.15 | 0.03 | 0.38 | 0.41 |
| SP500 ret | 0.05 | 0.03 | 0.06 | 0.07 | 0.79 |

Table 3: FEVD at 36 months (LOW vs HIGH)

**LOW**

| | IPGrowth | Inflation | FedFunds | BAA_AAA | SP500Ret |
|---|---|---|---|---|---|
| IP growth | 0.89 | 0.03 | 0.04 | 0.03 | 0.01 |
| Inflation | 0.02 | 0.82 | 0.05 | 0.04 | 0.06 |
| FedFunds | 0.15 | 0.17 | 0.54 | 0.03 | 0.11 |
| BAA–AAA | 0.06 | 0.31 | 0.12 | 0.44 | 0.06 |
| SP500 ret | 0.03 | 0.03 | 0.05 | 0.02 | 0.87 |

**HIGH**

| | IPGrowth | Inflation | FedFunds | BAA_AAA | SP500Ret |
|---|---|---|---|---|---|
| IP growth | 0.72 | 0.06 | 0.03 | 0.05 | 0.14 |
| Inflation | 0.07 | 0.62 | 0.16 | 0.11 | 0.04 |
| FedFunds | 0.04 | 0.20 | 0.58 | 0.15 | 0.03 |
| BAA–AAA | 0.03 | 0.15 | 0.03 | 0.38 | 0.41 |
| SP500 ret | 0.05 | 0.03 | 0.06 | 0.07 | 0.78 |

# References


[1] L. J. Christiano, M. Eichenbaum, and C. L. Evans. *Monetary Policy Shocks: What Have We Learned and to What End?* NBER Working Paper 6400, 1998.

[2] H. Uhlig. What are the effects of monetary policy on output? Results from an agnostic identification procedure. *Journal of Monetary Economics*, 52:381-419, 2005.